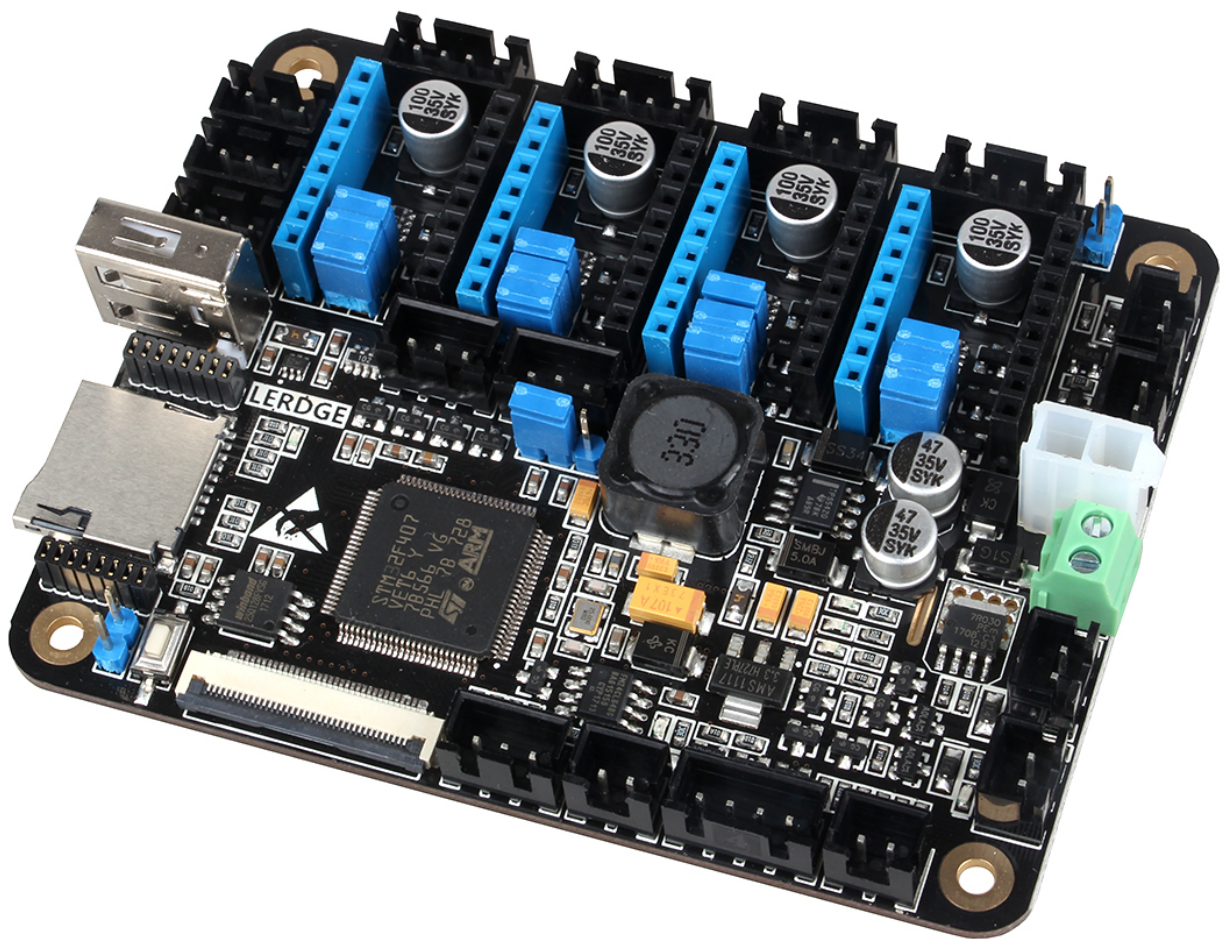


The Process and Precautions of Upgrading Bootloader for Lerdge-X Motherboard



Note:

1. Bootloader is not the executable firmware of the motherboard. Please do not confuse it with the motherboard firmware.
2. Before updating the bootloader, please be sure to distinguish the bootloader upgrade files of the Lerdge-S motherboard and the Lerdge-X motherboard to avoid upgrade errors.
3. Updating the bootloader may lose the parameters in the system. Please make a parameter backup in advance.
4. After updating the Bootloader, you need to program the firmware by forced update. Only the TF card can be used for forcing the update. Please prepare a TF card with a capacity less than 4G, and make sure that the TF card can be used for forced update. (If the capacity of the TF card is greater than 4G, please perform FAT32 format to clear the TF card content first.)

1. Prepare a TF card.
2. Unzip the Lerdge_X_Bootloader.rar archive to the location you can find. (eg extract to the desktop)
3. Copy the "Lerdge_X_Bootloader_V1.0.4.bin" file from the Lerdge_X_Bootloader folder to the location you can find in the TF card.
4. After the motherboard is powered on, click the "⚙️" icon to enter the "System" interface, click the "📁" icon, and select the "💾" icon, select the "Lerdge_X_Bootloader_V1.0.4.bin" file in the TF card and click the "OK" icon to perform the update process.
5. After the update progress bar is completed, the system will jump to the Boot loader upgrade interface. As shown in Figure 1

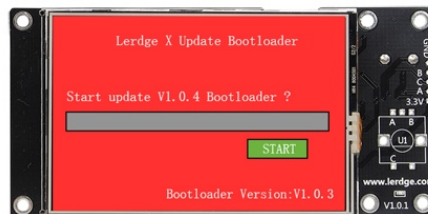


Figure 1

6. Click the "START" button to start upgrading the motherboard boot loader.
Do not power off during the upgrade process! Do not power off! Do not power off! Power failure will make the motherboard to brick! !! Also do not operate the motherboard until the upgrade is complete.
7. Wait for the upgrade progress. After the execution is completed, the upgrade interface becomes as shown in Figure 2.

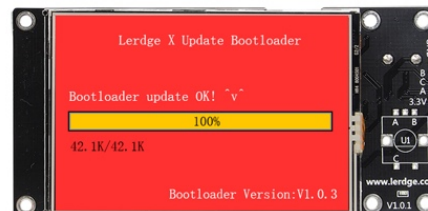


Figure 2

After the boot loader is upgraded, restarting the motherboard will not enter the execution program. The system still runs the Bootloader upgrade program. The system interface is displayed as shown in Figure 3.

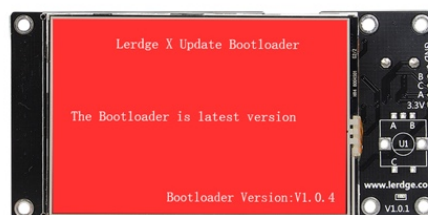


Figure 3

8. After the Bootloader upgrade is completed, there is no executable firmware in the system. You need to re-program the execution firmware to the motherboard by forcibly updating the firmware to use the motherboard normally.

About how to force firmware update, please refer to "**The Method and Precautions of Lerdge-X Motherboard Software Update**"